Lecture 12

Tuesday, 22 September 2026 7:20 PM

Correlation & Covariance

X, Y ind. $\Rightarrow \mathbb{E}(XY) = \mathbb{E}(X)\cdot\mathbb{E}(Y)$

$\mathbb{E}(XY)$ → Correlation b/w X, Y

$\mathbb{E}(XY) = 0$ → X, Y are said to be orthogonal.

$\mathbb{E}\left((X-\mu_X)(Y-\mu_Y)\right)$ → Covariance b/w X, Y

$\mathbb{E}(X)$ $\quad$ $\mathbb{E}(Y)$ $\quad$ $\mathrm{Cov}(X,Y)$

$$\mathrm{Cov}(X,Y) = \mathbb{E}\left[XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y\right]$$

$$= \mathbb{E}(XY) - \mu_X\cdot\mu_Y - \mu_Y\cdot\mu_X + \mu_X\mu_Y$$

$$= \mathbb{E}(XY) - \mu_X\cdot\mu_Y \quad \blacksquare$$

X, Y being uncorrelated $\iff \mathbb{E}(XY) = \mu_X\cdot\mu_Y$

$\iff \mathrm{Cov}(X,Y) = 0$

$\mathrm{Cov}(X,Y) = 0 \Rightarrow$ Independence if

① X, Y are jointly Gaussian

$x, y \in \mathbb{R}^2$ $\quad f_{XY}(x,y) = \dfrac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\dfrac{[x^2 - 2\rho xy + y^2]}{2(1-\rho^2)}\right)$

$\Rightarrow \mathrm{Cov}(X,Y) = 0 \Rightarrow \rho = 0$

$$f_{XY}(x,y) = \frac{1}{2\pi}\exp\left(-\frac{[x^2+y^2]}{2}\right)$$

$$= \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)\cdot\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{y^2}{2}\right)$$

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Independence

$f_X(x) \sim N(0,1)$ $\quad$ $f_Y(y) \sim N(0,1)$

② Bernoulli RVs $X, Y \in \{0,1\}$

$\forall\ x \in \{0,1\},\ y \in \{0,1\}$

$P_{XY}(x,y) = P_X(x) \cdot P_Y(y)$ ⟺ Ind.

$\Rightarrow E(XY) = E(X) \cdot E(Y)$

Eg $X \in \{-2, -1, 1, 2\}$ w.p $1/4$ each

$Y \in \{-1, 1\}$ w.p $1/2$ each

X, Y ind. $\quad Z = X \cdot Y$

① Are Z, X ind? X

② Are Z, X uncorrelated?

$\text{Cov}(Z, X) = 0$?

$E(Z \cdot X) \overset{?}{=} E(Z) \cdot E(X)$

$E(X^2 Y) \overset{?}{=} E(X^2) \cdot E(Y)$

$E(X^2 Y) = E(X^3) = 0$ $\quad$ $E(X^2) \cdot E(Y) = 0$

$E(X) = 0$

$P(X=1, Z=2) \overset{?}{\neq} P(X=1) \cdot P(Z=2)$

$X \cdot Y = 2$

$P(X=1, Z=2) = 0$ $\quad$ $P(X=1) > 0$ $\quad$ $P(Z=2) > 0$

$E[aX] = a\,E(X)$

$E(X+Y) = E(X) + E(Y)$

Properties of Variance $\quad \text{Var}(X) = E((X - \mu_X)^2)$

① $\text{Var}(aX) = a^2 \cdot \text{Var}(X)$

$\text{Var}(aX) = \text{Var}(Z) = E((Z - \mu_Z)^2) = E((aX - a\mu_X)^2)$

$E[Z] = E[aX] = a \cdot \mu_X$

$= a^2 \cdot E((X - \mu_X)^2)$

$= \text{Var}(X)$

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② $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$

if X, Y are uncorrelated

or

$\mathrm{Cov}(X,Y) = 0$

$\mathrm{Var}(X+Y)$

$= E\left((X+Y-(\mu_X+\mu_Y))^2\right)$

$= E\left((X-\mu_X)+(Y-\mu_Y))^2\right)$

$= \underbrace{E((X-\mu_X)^2)}_{\mathrm{Var}(X)} + E((Y-\mu_Y)^2) \} \to \mathrm{Var}(Y)$

$+ 2\,\underbrace{E((X-\mu_X)\cdot(Y-\mu_Y))}_{\mathrm{Cov}(X,Y)}$

$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + \underbrace{2\cdot\mathrm{Cov}(X,Y)}_{0 \text{ if } X, Y \text{ uncorrelated}}$

Generalize

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{Var}(X_i) \quad \checkmark$$

$$+ 2\cdot\sum_{i<j} \mathrm{Cov}(X_i, X_j) \quad \checkmark$$

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Eg – n balls, n bins

X = # empty bins

$E(X)$ $\quad X = \sum_{i=1}^{n} X_i$ $\rightsquigarrow$ 1 if Bin i empty, 0 o.w.

$$E(X) = \sum_{i=1}^{n} E(X_i) = n \cdot E(X_1)$$

$$= n \cdot P(\text{Bin } 1 \text{ is empty})$$

$$= n \cdot \left(\frac{n-1}{n}\right)^n$$

$\mathrm{Var}(X)$ $\quad X$ = # empty bins

$$X = \sum_{i=1}^{n} X_i$$

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \underbrace{\sum_{i=1}^{n} \mathrm{Var}(X_i)}_{(A)} + \underbrace{2 \cdot \sum_{i<j} \mathrm{Cov}(X_i, X_j)}_{(B)}$$

$\mathrm{Var}(X_1) = p(1-p)$ $\quad (A)$

(B) $\quad \left(\frac{n-1}{n}\right)^n \triangleq p$

$X_1 \sim \mathrm{Ber}\left(\left(\frac{n-1}{n}\right)^n\right) = \mathrm{Ber}(p)$

$$(A) = n \cdot p \cdot (1-p)$$

$$(B) = 2 \cdot \sum_{i<j} \mathrm{Cov}(X_i, X_j)$$

$$\mathrm{Cov}(X_i, X_j) = E(X_i \cdot X_j) - \underbrace{E(X_i)}_{p} \cdot \underbrace{E(X_j)}_{p}$$

$X_i \cdot X_j = 1$ if Bins i & j are both empty

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$$E(X_i \cdot X_j) = P(\text{Bins } i, j \text{ are both empty})$$

$$= \left(\frac{n-2}{n}\right)^n$$

$$\text{Cov}(X_i, X_j) = \left(\frac{n-2}{n}\right)^n - \left(\frac{n-1}{n}\right)^{2n}$$

$$\textcircled{B} = 2 \cdot \sum_{i<j} \text{Cov}(X_i, X_j) \qquad \binom{n}{2} = \frac{n(n-1)}{2}$$

$$= n(n-1) \cdot \left[\left(\frac{n-2}{n}\right)^n - \left(\frac{n-1}{n}\right)^{2n}\right]$$

$$\text{Var}(X) = \text{Var}(\#\text{ empty bins})$$

$$= \textcircled{A} + \textcircled{B}$$

$$= np(1-p) + n(n-1)\left[\left(\frac{n-2}{n}\right)^n - p^2\right]$$

Covariance

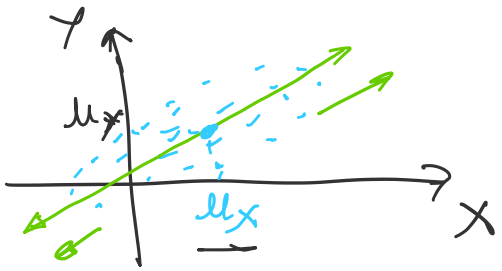
$$\text{Cov}(X, Y) = E((X - \mu_X)(Y - \mu_Y))$$

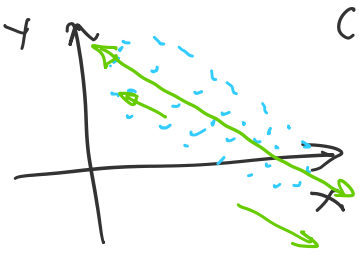
Measure of linear dep. b/w X, Y

+ve $\text{cov}(X, Y) \rightarrow$ X, Y grow together

-ve $\text{cov}(X, Y) \rightarrow$ X, Y grow opp.

(X, Y) +ve cov





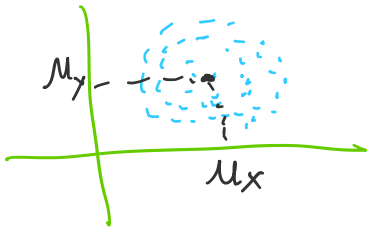
(X, Y) -ve cov

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$Y = aX + b$

$a \downarrow \frac{Cov(X,Y)}{Var(X)}$

$$E\left((X-\mu_X)(Y-\mu_Y)\right)$$



$Cov(X,Y) = 0$

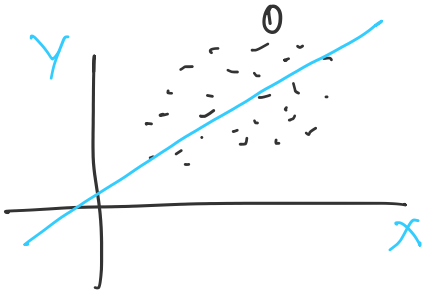
(1) $Cov(X,Y)$ only measures lin. dep.

$Cov(X,Y) = 0$ does not nec. mean no. dep.

Eg $X \in \{-1, 1\}$ w.p $1/2$ each. $Y = X^2$

$$Cov(X,Y) = E(X \cdot Y) - \mu_X \cdot \mu_Y$$
$$= E(X^3) - 0 \cdot = 0$$

$X^3 \to -1 \to 1/2$
$\searrow 1 \to 1/2$



$Cov(X,Y) = E\left((X-\mu_X)(Y-\mu_Y)\right)$

$Cov(bX, Y) = b \cdot Cov(X,Y)$ not invariant

$Cov(X,X) = Var(X)$

$E[(X-\mu_X)(X-\mu_X)]$

Correlation coefficient

$$\rho(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}}$$ [dimensionless]

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$\rho(X,Y) \in [-1, 1]$

$\rho(X,Y) \approx 1$ if Y is lin fn of X with +ve slope

$\rho(X,Y) \approx -1$ if Y is a lin. fn of X with -ve slope

$\rho(X,Y)$ $\boxed{\rho(bX, cY)}$

$$= \frac{\text{Cov}(bX, cY)}{\sqrt{\text{Var}(bX)} \cdot \sqrt{\text{Var}(cY)}}$$

$$= \frac{b \cdot c \cdot \text{Cov}(X,Y)}{b\sqrt{\text{Var}(X)} \cdot c\sqrt{\text{Var}(Y)}} = \rho(X,Y)$$

$\text{Cov}(X, Y) = \text{Cov}(Y, X)$

$\text{Cov}(X, X) = \text{Var}(X)$

$\text{Cov}(bX, cY) = b \cdot c \cdot \text{Cov}(X, Y)$

$$\text{Cov}\left(\sum_i X_i, \sum_j Y_j\right) = \sum_i \sum_j \text{Cov}(X_i, Y_j)$$

Eg Sales → Target $\boxed{Y = 2A + W}$

1. Adv. spend $X_1 = A$
2. Website traffic $X_2 = A + U$
3. Sales calls / inquiries. $X_3 = .5A + V$

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$A, U, V, W \rightarrow$ Independent

$\rightsquigarrow E(A) = E(U) = E(V) = E(W) = 0$

$\rightsquigarrow \text{var}(A) = \text{var}(U) = \text{var}(V) = \text{var}(W) = 1$

$X_1, X_2, X_3 \quad \ldots \quad Y$

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	X_1	X_2	X_3
X_1	.	ρ_{12}	ρ_{13}
X_2		.	
X_3			.

$$\rho_{12} = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{var}(X_1)}\sqrt{\text{var}(X_2)}}$$

$X_1 = A, \quad X_2 = A + U$

$\text{Var}(X_1) = \text{var}(A) = 1$

$\text{var}(X_2) = \text{var}(A+U) = \text{var}(A) + \text{var}(U)$

$= 2$

$\text{Cov}(X_1, X_2) = \text{Cov}(A, A+U)$

$= \text{Cov}(A, A) + \text{Cov}(A, U)$

$= \text{var}(A) + 0$

$= 1.$

$\text{Cov}(X_1, X_2) = 1$

$$\rho_{12} = \frac{1}{\sqrt{1} \cdot \sqrt{2}} = \frac{1}{\sqrt{2}} = .71$$

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	X_1	X_2	X_3
X_1	1	.71	.45
X_2	.71	1	.32
X_3	.45	.32	1

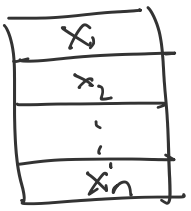
X_1, X_2 → have high positive correlation

$Y \quad \overleftarrow{X_1, X_2, X_3}$

$\rho_{YX_1} \qquad \rho_{YX_2}$

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100 100 100



G_1, G_2 with rep. bias

$\hat{X} = \frac{G_1 + G_2}{2}$

$E(\hat{X}) = \frac{(\sum x_i)}{n}$

$E\left((X - \mu_X)(Y - \mu_Y)\right) = E(X \cdot Y)$

$(x_3, y_3) \to x_3 \cdot y_3$
$(x_1, y_1) \quad x_1 \cdot y_1$
$(x_2, y_2) \to x_2 \cdot y_2$

$X \in \Sigma_X, \quad Y \in \Sigma_Y$

$\{0, 1, 2\}$ $\quad$ $\{0, 1\}$

$\frac{1}{3}$ $\quad$ $\frac{1}{2}$ $\frac{1}{2}$

$E(X \cdot Y)$

$0 \to P(0,0) + p(0,1) + p(1,0) + p(2,0)$

$1 \to p(1,1)$

$2 \to p(2,1)$

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